

Q1	A	B	C	D	E	F
Q2	A	B	C	D	E	F
Q3	A	B	C	D	E	F
Q4	A	B	C	D	E	F
Q5	A	B	C	D	E	F
Q6	A	B	C	D	E	F
Q7	A	B	C	D	E	F
Q8	A	B	C	D	E	F
Q9	A	B	C	D	E	F
Q10	A	B	C	D	E	F
Q11	A	B	C	D	E	F
Q12	A	B	C	D	E	F
Q13	A	B	C	D	E	F
Q14	A	B	C	D	E	F
Q15	A	B	C	D	E	F
Q16	A	B	C	D	E	F
Q17	A	B	C	D	E	F
Q18	A	B	C	D	E	F
Q19	A	B	C	D	E	F

1. Which object acts as the power source in a simple battery-operated circuit? [1]

A switch

A clothespin

An alarm buzzer

A light bulb

A battery

A piece of string

2. What do some switches insert between two wires to stop the flow of electricity? [1]

Connecting wires

Insulators

Conductors

Power sources

Extra batteries

Metal components

3. In the analogy of children holding hands in a circle, what do the hands and arms represent in an electrical circuit? [1]

The emergency signal

The power source

Wires connecting the parts

The device needing power

The insulators

The switch

4. Which of the following is a common daily use for an alarm? [1]

Keeping a circuit perfectly separated

Insulating rubber from a power source

Changing the battery in a flashlight

Reminding you to take dinner out of the oven

Heating up toast for breakfast

Turning off a bedroom light

5. What must happen to a circuit for a connected light bulb to turn on? [1]

All parts must be wired together in a continuous, connected loop

The power source must be disconnected

The wires must be separated by an insulator

A buzzer must be added to the wiring

A switch must separate the two main wires

The battery must be placed outside the loop

6. During the redesign process, what is typically done immediately after deciding what changes to make and drawing the new design? [1]

Hiding the alarm materials

Purchasing new treasures

Revising or re-creating the model

Brainstorming initial concepts

Showing it to the entire class

Evaluating the initial results

7. What is the primary function of a power source in a circuit? [1]

To break the loop of electricity

To sound a loud buzzer during an emergency

To separate two electrical components securely

To keep valuable items safe from intruders

To act as a protective layer over wires

To provide the energy needed for devices to work

8. What is an electrical circuit? [1]

A piece of rubber used to block electricity

A connected loop that allows power to flow

A type of battery used in households

A switch that automatically controls lights

An unconnected wire dangling from a ceiling

A device used strictly for sounding emergencies

9. What can trigger a safety alarm in certain vehicles? [1]

Drifting into another lane by accident

Keeping valuable items safe

Disconnecting the battery

Turning on the radio

Using the windshield wipers

Turning on the bedroom light

10. What term describes materials that do not allow electricity to flow? [1]

Buzzers

Power sources	The buzzer disconnects from the wall
Conductors	The battery drops onto the floor to break the circuit
Connectors	The alarm gets blocked by an insulator
Circuits	Opening the door flips a glass switch
Insulators	The door cuts the wires with a hidden blade
11. When an alarm is triggered by smoke inside a building, what is its main function? [1]	A moving wire touches another wire to complete the circuit
To test the wire cutters	14. What is the first step in the formal design process? [1]
To release insulators into the air	Triggering the alarm buzzer
To signal an emergency	Evaluating the flaws
To interrupt the main power grid	Building the final model
To shut off the circuit instantly	Testing the prototype
To act as a power source	Describing, drawing the design, and listing materials
12. What is the primary purpose of an alarm? [1]	Revising the broken components
To create new loops in electrical circuits	15. Which of the following is considered a good insulator? [1]
To teach visual literacy in classrooms	A metal paperclip
To keep people or valuable items safe	A metal brad
To replace standard light bulbs	Copper wire
To provide energy to a battery	Glass
To block the flow of electricity indefinitely	A silver coin
13. How does a simple door alarm circuit function when the door opens? [1]	Aluminum foil

16. What happens when a circuit is interrupted? [1]
- The wires become permanently damaged
- A switch immediately changes the direction of flow
- The alarm buzzer gets significantly louder
- The power source doubles its output
- The circuit automatically redesigns itself
- The power stops flowing and the device turns off**
17. What is the testable goal when creating a protective alarm for a box? [1]
- The wires break off once the box is lifted
- The battery recharges when the lid is shut
- The alarm buzzes when someone opens the box**
- The box automatically locks itself permanently
- The box changes color when handled
- The alarm produces a bright flash of light
18. Which component is typically built in by engineers to easily connect or separate two wires? [1]
- A plastic tube
- A switch**
- A piece of string
- A glass pane
- A metal brad
- A rubber band
19. What are the standard sequential steps in the basic design process? [1]
- Plan, Create, Test, Evaluate**
- Test, Plan, Evaluate, Redesign
- Plan, Evaluate, Create, Test
- Create, Redesign, Revise, Test
- Evaluate, Create, Plan, Test
- Create, Test, Plan, Evaluate